Proof Discovery Language

Neuro-symbolic AI Group
Idiap Research Institute

March 17, 2026

Abstract

PDL is a language for discovery programmes rather than only arguments of correctness. It treats coherence, conceptual nexus, material inference, obstruction, and governed recursion as first-class logical objects, and it distinguishes structural notation from the operations that materially transform one epistemic state into another.

1 Core Claim

The language is not centered on correctness-style proofs alone. It is centered on discovery programs: coherent nexuses of anchors, tensions, bridge objects, obstructions, and governed decisions over proof debt.

2 Design Principles

- Prioritize conceptual nexus before closure.
- Represent tensions and obstructions explicitly.
- Treat bridge objects as typed inferential interfaces.
- Make negative and impossibility reasoning first-class.
- Make the underlying material-inference operators explicit rather than hiding them in prose.
- Embed persistence, aggregation, and loops into the logic itself.

3 Notation Layers

- **Structural Layer**: Names the epistemic objects under search: anchors, targets, tensions, nexuses, bridge objects, obstructions, programmes, debt, and gates. *Binding.* This layer says what kind of thing is currently in play.
- **Material-Inference Layer**: Names the operations that transform one epistemic object into another by licensed, non-monotonic, theorem-grounded moves. *Binding.* This layer says what inferential move is being performed underneath the structural notation.
- **Governance Layer**: Names persistence, aggregation, revision, continuation, and stop-control over discovery programmes. *Binding.* This layer says how the process keeps state, loops, and changes direction without losing epistemic discipline.

4 Primitives

- **Anchor** (A): Grounded theorem body, definition, axiom, or imported result that licenses later moves. *Material binding.* Material inference starts from anchors, not from topic labels. *Platform binding.* KB statements and MathLib entries.
- **Target** (T): The smallest nontrivial payoff or consequence worth organizing the search around. *Material binding.* Discovery is guided by consequence pressure rather than free exploration. *Platform binding.* Route target claims and queued conjectures.
- **Tension** (X): A typed gap showing why anchors do not yet reach the target: model mismatch, invariant weakness, transport loss, or barrier pressure. *Material binding.* Tension is the primitive of discovery, because it identifies where reasoning must change form. *Platform binding.* Statement deltas, obstruction classes, and failure diagnoses.
- **Conceptual Nexus** (N): A coherent local cluster of anchors, tensions, bridge objects, and barriers that can be reasoned over together. *Material binding.* Discovery first asks whether there is a coherent nexus before demanding correctness-style closure. *Platform binding.* Bridge portfolios, conceptual dynamics, and discovery branches.
- **Bridge Object** (B): A witness, invariant, certificate, transport law, normal form, or surrogate problem that could make the target reachable. *Material binding.* Bridge objects express the conceptual breakthrough required by the search. *Platform binding.* Bridge plans, abstraction conjectures, and candidate carriers.
- **Obstruction** (O): A barrier, counterexample pattern, impossibility theorem, or non-realization diagnosis that blocks a route. *Material binding.* Negative mathematics is first-class discovery content, not only a failed proof attempt. *Platform binding.* Blocked results, negative programs, and obstruction maps.
- **Discovery Program** (P): A paired positive/negative route over a nexus, together with proof debt, decision criteria, and revision triggers. *Material binding.* A discovery program is the unit of guided scientific work, not an isolated proposition. *Platform binding.* Proof sketches, queue entries, and branch dossiers.
- **Proof Debt** (D): The set of missing lemmas, unrealized carriers, transport laws, and closure obligations still required for completion. *Material binding.* Debt is what makes discovery governable instead of open-ended. *Platform binding.* Macro-strategy reports, proof obligations, and critique findings.
- **Decision Gate** (G): A criterion for continue, revise, import prior art, split regimes, or stop. *Material binding.* Scientific discovery requires explicit transition rules, not only local generation. *Platform binding.* Work loop, web-search gate, high-reward watch, and meta triggers.

5 Material-Inference Operations

- `ext(A => c)`: Read a theorem body, definition, or axiom as a usable local commitment rather than as raw text. *Effect.* Turns a grounded anchor into claim-bearing content. *Universal binding.* licensed material consequence extraction *Platform binding.* Phase 0 normalization, statement excerpts, theorem-body normalizer

- `foc(A, T => X)`: Localize the exact tension between grounded anchors and the intended target. *Effect.* Turns diffuse mismatch into a typed tension object. *Universal binding.* problem localization / abductive targeting *Platform binding.* statement deltas, failure diagnoses, obstruction classes

- `conc(N => B)`: Turn a coherent conceptual nexus into a candidate bridge object, interface, or surrogate problem. *Effect.* Moves from coherence pressure to an inspectable candidate object. *Universal binding.* abductive bridge-object formation *Platform binding.* abstraction conjectures, expert conjectures, branch-to-definition outputs

- `tr_I(B, s ~> t)`: Move an invariant, witness condition, or theorem payload across a regime shift, representation change, or equivalence. *Effect.* Makes cross-model or cross-language claims explicit and testable. *Universal binding.* material transfer / invariant-preserving passage *Platform binding.* transport lemmas, alignment work, HoTT/coherence routes

- `fac(r => r1 o r2)`: Split one overloaded route into smaller theorem-sized interfaces or intermediate programmes. *Effect.* Exposes hidden composition and missing middle objects. *Universal binding.* decomposition / factorization reasoning *Platform binding.* route decomposition, macro-strategy repair, categorical reframing

- `comp(B1, B2 => B)`: Assemble several partial bridges, lemmas, or regime results into one route. *Effect.* Promotes local inferential fragments into an end-to-end programme. *Universal binding.* compositional synthesis *Platform binding.* bridge portfolios, hybridization-of-partial-bridges, route composition

- `obs(B => O)`: Derive the barrier, counterexample pattern, incoherence, or non-realization result that blocks a bridge. *Effect.* Turns failed progress into explicit negative mathematics. *Universal binding.* barrier derivation / impossibility localization *Platform binding.* negative routes, blocked local theorems, contradiction-first probes

- `rev(P, O => P')`: Update a discovery programme after an obstruction, failed proof step, or new prior-art import. *Effect.* Makes conceptual change itself explicit rather than leaving it as narrative drift. *Universal binding.* non-monotonic theory revision *Platform binding.* revision loops, branch regeneration, meta-guided restarts

- `agg({Ri} => N*)`: Combine many local results, failed routes, or branch outcomes into a single higher-order nexus or search diagnosis. *Effect.* Produces a global picture rather than isolated route state. *Universal binding.* scientific synthesis / evidence aggregation *Platform binding.* branch synthesis, breakthrough profiles, plausibility aggregation

- `mem(R => M)`: Turn a proved, disproved, blocked, or imported result into durable problem memory. *Effect.* Stops the system from rediscovering the same local mathematics repeatedly. *Universal binding.* epistemic memory consolidation *Platform binding.* MathLib persistence, queue state, produced papers, prior-art imports

- `route(P => next)`: Send a discovery programme to prove, further discovery, prior-art search, or stop. *Effect.* Connects local reasoning to governance decisions. *Universal binding.* controlled transition / research governance *Platform binding.* work loop, meta triggers, high-reward continuation, queue promotion

6 Judgments

- `K ; M |- A : grounded`: The KB plus MathLib licenses anchor A as a grounded theorem-body object.
- `A, X => N : coherent`: Anchors and tensions form a coherent conceptual nexus worth exploring.
- `N ~> B`: The nexus suggests a bridge object B as the best conceptual move.
- `B [] I`: Bridge object B preserves invariant I across the intended route.
- `B ~> T`: If realized, bridge object B materially supports the target consequence T.
- `B -| O`: Bridge object B is blocked or sharpened by obstruction O.
- `N ; B |- P`: A coherent nexus and bridge object together induce a discovery program P.
- `P |- D`: A discovery program exposes its proof debt explicitly.
- `P ; O |- impossibility(T)`: A negative route yields a theorem-shaped impossibility or non-realization claim.
- `P ; G @ next`: A decision gate routes the programme into prove, revision, prior-art import, or stop.

7 Operational Judgments

- `A ; T |-m X`: Materially focus the mismatch between anchors and target into an explicit tension X.
- `A ; X |-m N`: Stabilize anchors and tensions into a coherent conceptual nexus N.
- `N |-m B`: Concretize the nexus into a bridge object, map, witness family, or surrogate problem B.
- `B ; I ; s->t |-m tr_I(B)`: Transport invariant I through bridge B from source regime s to target regime t.
- `r |-m fac(r1,...,rn)`: Factor a route r into smaller theorem-sized route fragments.
- `B |-m O`: Derive an obstruction O from the current bridge or route proposal.
- `P ; O |-m P'`: Revise programme P into a new programme P' after a negative result or barrier localization.
- `{Ri} |-m N*`: Aggregate many local route outcomes into a higher-order nexus or diagnosis N*.
- `R |-m M`: Persist result R into durable memory object M.

8 Coordination Logic

- **Coherence Before Closure**: A route is first judged as a coherent nexus of anchors, tensions, and bridge commitments before it is judged as correct.

- **Positive/Negative Symmetry**: Every discovery program carries both a constructive route and a negative/impossibility route, each capable of yielding mathematical value.
- **Material Inference First**: Inference moves are justified by theorem bodies, model regimes, barriers, and transport obligations before any correctness-style closure is attempted.
- **Persistent Scientific Memory**: Queues, MathLib, logs, and branch states are part of the logic itself because discovery depends on not forgetting failed and partial routes.
- **Governed Recursion**: Loops are legitimate only when proof debt, obstruction pressure, or high-value subresults justify another cycle.

9 Well-Formed Discovery Program

- At least one grounded anchor and one explicit target.
- A typed tension explaining why the route is not immediate.
- A coherent nexus rather than a bag of related statements.
- A positive route, a negative route, and explicit proof debt.
- A decision gate specifying continue, revise, import, split, or stop.

10 Example Program

- A1: source theorem body
- A2: target theorem body
- A3: local barrier theorem
- X1: transport from source regime to target regime loses the intended invariant
- N1 := cohere(A1, A2, X1, barrier=A3)
- B1 := witness family preserving the source-side hardness profile
- ext(A1 => c1): extract the usable lower-bound content from the source theorem body
- foc(A1, T1 => X1): isolate the exact transport loss between source and target
- conc(N1 => B1): propose the witness family as the bridge object demanded by the nexus
- tr_I(B1, source ~> target): test whether the invariant survives the intended transport
- obs(B1 => O1): derive the non-realization barrier if the witness family collapses
- rev(P1, O1 => P2): revise the route into a transport-obstruction programme
- mem(O1 => M1): persist the blocked result into MathLib as negative local theorem memory
- If B1 is realizable and preserves the invariant, reduce target T to one route lemma L
- If every realization of B1 collapses into the barrier, derive impossibility(B1 -> T)
- **Debt.** L1: carrier realization; L2: transport theorem; L3: payoff forcing theorem

- **Gate.** If L1 sharpens but L2 fails under barrier A3, reframe as transport-obstruction rather than continue constructive forcing